Weekly Progress Report 1

1. Dataset Loading

The dataset was loaded using the Pandas library.

```python
import numpy as np
import pandas as pd

#Load datasets
train = pd.read_csv('train.csv')
test = pd.read_csv('test.csv')
```

Two files were provided:

- train.csv
- test.csv

2. Understanding the Dataset

We used several Pandas functions to understand the dataset structure.

head()

Displays the first five rows of the dataset.

Output:

Train.csv

```
   id  Age  Sex  ...  Number of vessels fluro  Thallium  Heart Disease
0   0   58    1  ...                        2         7       Presence
1   1   52    1  ...                        0         3        Absence
2   2   56    0  ...                        0         3        Absence
3   3   44    0  ...                        0         3        Absence
4   4   58    1  ...                        3         3       Presence

[5 rows x 15 columns]

Process finished with exit code 0
```

Test.csv

```
       id  Age  Sex  ...  Slope of ST  Number of vessels fluro  Thallium
0  630000   58    1  ...            2                        3         3
1  630001   55    0  ...            1                        0         3
2  630002   54    1  ...            2                        3         7
3  630003   44    0  ...            1                        0         3
4  630004   43    1  ...            2                        0         7

[5 rows x 14 columns]

Process finished with exit code 0
```

shape()

Displays number of rows and columns.

Output:

Train.csv

```
(630000, 15)

Process finished with exit code 0
```

Test.csv

```
(270000, 14)

Process finished with exit code 0
```

info()

Shows:

- column names
- number of non-null values
- data types

This helps us check if any columns contain missing values

Output:

Train.csv

```
<class 'pandas.DataFrame'>
RangeIndex: 630000 entries, 0 to 629999
Data columns (total 15 columns):
 #   Column                   Non-Null Count   Dtype
---  ------                   --------------   -----
 0   id                       630000 non-null  int64
 1   Age                      630000 non-null  int64
 2   Sex                      630000 non-null  int64
 3   Chest pain type          630000 non-null  int64
 4   BP                       630000 non-null  int64
 5   Cholesterol              630000 non-null  int64
 6   FBS over 120             630000 non-null  int64
 7   EKG results              630000 non-null  int64
 8   Max HR                   630000 non-null  int64
 9   Exercise angina          630000 non-null  int64
 10  ST depression            630000 non-null  float64
 11  Slope of ST              630000 non-null  int64
 12  Number of vessels fluro  630000 non-null  int64
 13  Thallium                 630000 non-null  int64
 14  Heart Disease            630000 non-null  str
dtypes: float64(1), int64(13), str(1)
memory usage: 72.1 MB
None
```

Test.csv

```
<class 'pandas.DataFrame'>
RangeIndex: 270000 entries, 0 to 269999
Data columns (total 14 columns):
 #   Column                   Non-Null Count   Dtype
---  ------                   --------------   -----
 0   id                       270000 non-null  int64
 1   Age                      270000 non-null  int64
 2   Sex                      270000 non-null  int64
 3   Chest pain type          270000 non-null  int64
 4   BP                       270000 non-null  int64
 5   Cholesterol              270000 non-null  int64
 6   FBS over 120             270000 non-null  int64
 7   EKG results              270000 non-null  int64
 8   Max HR                   270000 non-null  int64
 9   Exercise angina          270000 non-null  int64
 10  ST depression            270000 non-null  float64
 11  Slope of ST              270000 non-null  int64
 12  Number of vessels fluro  270000 non-null  int64
 13  Thallium                 270000 non-null  int64
dtypes: float64(1), int64(13)
memory usage: 28.8 MB
None
```

3. Missing Values Analysis

```
# Finding the sum of missing values
print(train.isnull().sum())
```

```
print(test.isnull().sum())
```

This command checks if any columns contain missing values.

train.isnull().sum()

- **train.isnull()**: This method iterates through the DataFrame and creates a new DataFrame of the same size, where each cell contains a Boolean value.
- **True** indicates a missing value, and **False** indicates a non-missing value.
- **.sum()**: This method counts all the **True values** in the DataFrame. By default, it calculates the sum **column by column**, which shows the number of missing values in each column.

Output:

Train.csv:

```
id                        0
Age                       0
Sex                       0
Chest pain type           0
BP                        0
Cholesterol               0
FBS over 120              0
EKG results               0
Max HR                    0
Exercise angina           0
ST depression             0
Slope of ST               0
Number of vessels fluro   0
Thallium                  0
Heart Disease             0
dtype: int64

Process finished with exit code 0
```

Test.csv:

```
id                        0
Age                       0
Sex                       0
Chest pain type           0
BP                        0
Cholesterol               0
FBS over 120              0
EKG results               0
Max HR                    0
Exercise angina           0
ST depression             0
Slope of ST               0
Number of vessels fluro   0
Thallium                  0
dtype: int64

Process finished with exit code 0
```

4. Duplicate Records

```
# Duplicate row
print(train.duplicated().any())
```

```
print(test.duplicated().any())
```

This checks if any rows are repeated.

Output:

Train.csv

```
False

Process finished with exit code 0
```

Test.csv

```
False

Process finished with exit code 0
```

5. Statistical Summary

```python
# Statistical summary of all columns regardless of type
print(train.describe(include='all'))
```

```python
print(test.describe(include='all'))
```

This provides statistics for all columns regardless of type such as:

- mean
- minimum
- maximum
- standard deviation

Output:

Train.csv

```
                   id            Age  ...       Thallium  Heart Disease
count   630000.000000  630000.000000  ...  630000.000000         630000
unique            NaN            NaN  ...            NaN              2
top               NaN            NaN  ...            NaN        Absence
freq              NaN            NaN  ...            NaN         347546
mean    314999.500000      54.136706  ...       4.618873            NaN
std     181865.479132       8.256301  ...       1.950007            NaN
min          0.000000      29.000000  ...       3.000000            NaN
25%     157499.750000      48.000000  ...       3.000000            NaN
50%     314999.500000      54.000000  ...       3.000000            NaN
75%     472499.250000      60.000000  ...       7.000000            NaN
max     629999.000000      77.000000  ...       7.000000            NaN

[11 rows x 15 columns]

Process finished with exit code 0
```

Test.csv

```
                  id            Age  ...  Number of vessels fluro       Thallium
count  270000.000000  270000.000000  ...             270000.00000  270000.000000
mean   764999.500000      54.159870  ...                  0.45480       4.619774
std     77942.430678       8.255471  ...                  0.80127       1.950273
min    630000.000000      29.000000  ...                  0.00000       3.000000
25%    697499.750000      48.000000  ...                  0.00000       3.000000
50%    764999.500000      54.000000  ...                  0.00000       3.000000
75%    832499.250000      60.000000  ...                  1.00000       7.000000
max    899999.000000      77.000000  ...                  3.00000       7.000000

[8 rows x 14 columns]

Process finished with exit code 0
```